Appendix 22

The three-dimensional electron gas at high magnetic field

In this appendix we summarize some essential facts pertaining to the energetics of the *three-dimensional* electron liquid at high magnetic field. The one-electron energy subbands, given by Eq. (10.19), are plotted vs k_z in Fig. A22.1(a). We have introduced a Zeeman splitting $\Delta = \frac{g}{2}\hbar\omega_c$ between up-spin ($\sigma = +1$) and down-spin ($\sigma = -1$) states. The zero of the energy is set at the bottom of the lowest down-spin subband. Two dimensionless parameters characterize this system, namely, the usual electron gas parameter r_s and the ratio of the magnetic length ℓ to the Bohr radius a_B:

$$\frac{\ell}{a_B} = \sqrt{\frac{2}{\hbar\omega_c/\mathrm{Ry}}} \,. \tag{A22.1}$$

A22.1 Noninteracting kinetic energy

Let $\bar{\epsilon}_F \equiv \frac{\epsilon_F}{\hbar\omega_c}$ be the Fermi energy in units of the cyclotron energy. The j-th occupied subband crosses the Fermi level at wave vectors $k_z = \pm k_{j\sigma}$, where

$$k_{j\sigma}\ell = \sqrt{2\left(\bar{\epsilon}_F - j - g\frac{\sigma+1}{4}\right)} \,. \tag{A22.2}$$

It is convenient to set $k_{j\sigma} = 0$ for the unoccupied subbands, so that the three-dimensional electron density n is simply given by

$$n = \frac{1}{2\pi^2\ell^2}\sum_{j\sigma} k_{j\sigma} \,. \tag{A22.3}$$

Notice that, even though the sum over j is formally unrestricted, our definition of $k_{j\sigma}$ guarantees that only the occupied subbands contribute to the density. Eq. (A22.3) can be inverted numerically (see Fig. A22.1(b)), yielding the Fermi energy as a function of n. Once $\bar{\epsilon}_F$ is known, the non-interacting kinetic energy per particle, ϵ_0, is obtained from

$$\epsilon_0(n, B) = \frac{\hbar\omega_c}{6\pi^2 n}\sum_{j\sigma} k_{j\sigma}^3 \,. \tag{A22.4}$$

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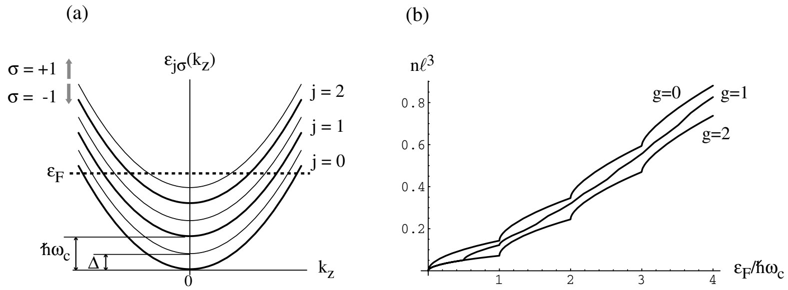

Fig. A22.1. (a) Landau level subbands for a three-dimensional electron gas with $g = 1$ in a magnetic field. Thick and thin lines correspond to down-spin and up-spin dispersions respectively. (b) Electronic density vs Fermi energy at three different values of the g-factor.

A22.2 Density–density response function

The calculation of the noninteracting density–density response function presented in Section 10.4.5 is easily extended to three dimensions, yielding

$$\chi_0(\vec{q},\omega) = -\frac{1}{4\pi^2\hbar\omega_c\ell^3}\sum_{jj'\sigma}\frac{|F_{j'j}(\vec{q}_\perp)|^2}{q_z\ell}\ln\left\{\frac{\omega^2-\left[\omega_{jj'}-\frac{\hbar q_z^2}{2m_b}-\frac{\hbar k_{j\sigma}q_z}{m_b}\right]^2}{\omega^2-\left[\omega_{jj'}-\frac{\hbar q_z^2}{2m_b}+\frac{\hbar k_{j\sigma}q_z}{m_b}\right]^2}\right\}, \tag{A22.5}$$

where $\omega_{jj'} \equiv (j-j')\omega_c$ and $\vec{q}_\perp$ is the component of $\vec{q}$ in the plane perpendicular to the magnetic field. The form factors $F_{j'j}(\vec{q}_\perp)$ are defined in Eq. (10.84).

A22.3 Noninteracting structure factor

The noninteracting static structure factor can be calculated directly from the fluctuation-dissipation theorem as follows:

$$\begin{aligned} S_0(\vec{q}) &= -\frac{\hbar}{\pi n}\int_0^\infty \chi_0(\vec{q},i\omega)d\omega \\ &= \frac{1}{4\pi^2 n\ell^3}\sum_{jj'\sigma}\frac{|F_{j'j}(\vec{q}_\perp)|^2}{q_z\ell}\left[\left|\frac{q_z^2\ell^2}{2}+k_{j\sigma}q_z\ell^2+j'-j\right|\right. \\ &\quad \left.-\left|\frac{q_z^2\ell^2}{2}-k_{j\sigma}q_z\ell^2+j'-j\right|\right], \end{aligned} \tag{A22.6}$$

where use has been made of the integral $\int_0^\infty \ln\frac{\omega^2+a^2}{\omega^2+b^2}d\omega = \pi(|a|-|b|)$. Because $S_0(\vec{q})$ is even under inversion of q_z there is no loss of generality in assuming $q_z > 0$. Evaluating the difference of the absolute values in Eq. (A22.6) leads to three different results, depending on the relative values of q_z, j, and j': the three regimes are

$$|\ldots|-|\ldots| = \begin{cases} \text{(i)}\ \ 2k_{j\sigma}q_z\ell^2\,, & q_z \geq k_{j\sigma}+k_{j'\sigma}\,, \\ \text{(ii)}\ \ q_z^2\ell^2+2(j'-j)\,, & |k_{j\sigma}-k_{j'\sigma}| < q_z < k_{j\sigma}+k_{j'\sigma}, \\ \text{(iii)}\ 2k_{j\sigma}q_z\ell^2\ \mathrm{sign}(j'-j)\,, & 0 < q_z \leq |k_{j\sigma}-k_{j'\sigma}|\,. \end{cases} \tag{A22.7}$$

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Notice that for $q_z > 2k_{0\downarrow}$ we are always in case (i) and it is easy to verify that $S_0(\vec{q}) = 1$. We can now take advantage of the fact that the sum over j and j' in Eq. (A22.6) is formally unrestricted to further simplify the summand: the term $2(j' - j)$ in case (ii) vanishes upon summation over j and j'; in (i) $2k_{j\sigma}$ can be replaced by $k_{j\sigma} + k_{j'\sigma}$, and in (iii) $2k_{j\sigma}\mathrm{sign}(j' - j)$ is replaced by $(k_{j\sigma} - k_{j'\sigma})\mathrm{sign}(j' - j) = |k_{j\sigma} - k_{j'\sigma}|$. Thus, after a few simple manipulations one arrives at

$$S_0(\vec{q}) - 1 \;=\; \frac{1}{4\pi^2 n\ell^2} \sum_{jj'\sigma} |F_{j'j}(\vec{q}_\perp)|^2\, \Theta_{j'j_\sigma}(|q_z|) \tag{A22.8}$$

where

$$\Theta_{j'j_\sigma}(|q_z|) = \begin{cases} 0\,, & |q_z| \geq k_{j\sigma} + k_{j'\sigma}\,, \\ |q_z| - k_{j\sigma} - k_{j'\sigma}\;, & |k_{j\sigma} - k_{j'\sigma}| < |q_z| < k_{j\sigma} + k_{j'\sigma}\;, \\ |k_{j\sigma} - k_{j'\sigma}| - k_{j\sigma} - k_{j'\sigma}\;, & |q_z| \leq |k_{j\sigma} - k_{j'\sigma}|\;. \end{cases} \tag{A22.9}$$

Notice that only the occupied subbands contribute to the static structure factor.

A22.4 Exchange energy

The exchange energy per particle is given by

$$\epsilon_x(n, B) \;=\; \frac{1}{2} \int \frac{d^3q}{(2\pi)^3} \frac{4\pi e^2}{q^2} \left[S_0(\vec{q}) - 1\right]\;. \tag{A22.10}$$

Substituting in this expression the non-interacting structure factor $S_0(\vec{q})$, and performing the integral over q_z analytically we arrive at a one-dimensional integral, which can be evaluated numerically:

$$\epsilon_x(n, B) = \frac{e^2}{4\pi^2 n\ell^2} \sum_{jj'\sigma} \int_0^\infty dq_\perp\, q_\perp |F_{j'j}(\vec{q}_\perp)|^2\, H_{j'j_\sigma}(q_\perp)\,, \tag{A22.11}$$

where $H_{j'j_\sigma}(q_\perp) = \int_0^\infty dq_z \frac{\Theta_{j'j_\sigma}(q_z)}{q_\perp^2 + q_z^2}$.

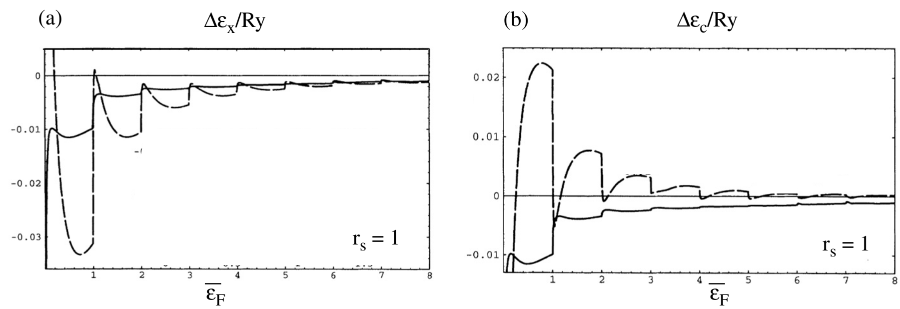

Fig. A22.2. (a) Dashed line: the difference $\Delta\epsilon_x(n, B) \equiv \epsilon_x(n, B) - \epsilon_x(n, 0)$ vs $\bar{\epsilon}_F = \frac{\epsilon_F}{\hbar\omega_c}$ at constant density ($r_s = 1$), for vanishing Zeeman splitting ($g = 0$). (b) Dashed line: the difference $\Delta\epsilon_c(n, B) \equiv \epsilon_c(n, B) - \epsilon_c(n, 0)$ vs $\bar{\epsilon}_F$ at constant density ($r_s = 1$) for $g = 0$. In both panels the solid line represents the difference $\Delta\epsilon_{xc}(n, B) \equiv \epsilon_{xc}(n, B) - \epsilon_{xc}(n, 0)$, where $\epsilon_{xc}(n, B)$ is the exchange-correlation energy. (Adapted from Skudlarski and Vignale, 1993.)

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In Fig. A22.2(a) we show how the exchange energy evolves as a function of magnetic field at constant density. Notice that the exchange energy is generally more negative than that of the electron liquid at the same density in zero magnetic field.

A22.5 Correlation energy

The correlation energy in RPA is straightforwardly calculated from Eq. (10.91) (Skudlarski and Vignale, 1993): its evolution as a function of magnetic field is plotted in Fig. A22.2(b). Unlike the exchange energy, the RPA correlation energy is generally less negative than in the electron liquid at the same density in zero magnetic field. The full exchange-correlation energy, however, is always rendered more negative by the magnetic field. These qualitative features have been recently confirmed by Takada and Goto (1998). These authors have calculated the correlation energy in the STLS approximation (see Section 5.4.4) and have also provided plots of the STLS local field factor $G_+(q)$ in the presence of a magnetic field. Analytical fits for the exchange-correlation energy vs density and magnetic field can be found in the cited papers.